

- **Title:** Exploring the Fundamentals of Information Theory through Musical Data: An Introductory Workshop in HumdrumR
- **Author:** Nat Condit-Schultz (Georgia Institute of Technology)



HumdrumR is an R package for analyzing data encoded in the [humdrum data syntax](#), a powerful and flexible format for musical data with many large datasets available online. Inspired by the design of the original humdrum toolkit (Huron 1995), humdrumR is designed to be appealing and accessible to traditional musicologists while also being useful to more experienced computational researchers. HumdrumR combines the flexibility and transparency of the humdrum syntax with the powerful data analysis tools and concise syntax of R, making basic musicological analyses (i.e., tallying scale degrees) extremely simple and easy while simultaneously affording complex analyses, statistical testing, and graphing. HumdrumR also includes a suite of tools for conducting analyses based on [information theory](#), including new (as of 2026) tools for modeling the information dynamics of music using the *prediction by partial matching* algorithm (Cleary & Witten, 1984), just like the [IDyOM](#) (Pearce, 2005) system.

This workshop is an introduction to humdrumR, suitable for music scholars with little or no experience with either humdrum or R. We will focus on the analysis of pitch and rhythm data in several well-known humdrum datasets: the [Bach 370 Chorales](#), the [Essen folk-song collection](#), [TAVERN](#), and [CoCoPops](#). Though we will proceed at the pace of the workshop participants and explore any topics/questions of interest to them, the focus of the workshop will be the application of ideas from **information theory** to musical data. We will show how to compute—and interpret—basic information-theory metrics (entropy, mutual information) across and between different representations of pitch and rhythm, and how to compare these metrics between different (sub)sets of musical data, or across time. We will take time to discuss and *understand* what information-theory metrics actually mean, and how they can be meaningful tools for musicological analysis—particularly but not limited to their connection to the perception and cognition of music.

The planned (but flexible) schedule of topics is:

1. Tabulating pitch data in various forms (generic/simple pitch, absolute pitch vs scale degrees, etc.).
2. Computing (and interpreting) entropy and information theory.
3. Comparing distributions and entropy between composers/genres, and across time.
4. Exploring correlations and “mutual information” between rhythm and pitch.
5. Simple N-gram tabulation, analyses, and conditional entropy.
6. Complex N-gram analysis: Information Dynamics of Music (“IDyOM”-type computation).

Workshop participants will need to bring their own computers. You do not need to install or download anything in advance, as all workshop materials (and a functioning R instance) will be running on the cloud (Posit cloud). If you *wish* to install humdrumR or download datasets on your machine in advance, detailed instructions will be provided in advance of the meeting.